
KAULIV Solution for Injection

NAME OF THE PRODUCT:

KAULIV 20 micrograms/80 microliters solution for injection

QUALITATIVE AND QUANTITATIVE COMPOSITION:

Each dose of 80 microliters contains 20 micrograms of teriparatide*.

Each cartridge of 3 mL of solution contains 750 micrograms of teriparatide (corresponding to 250 micrograms per mL).

*Teriparatide, rhPTH (1-34), produced in *E. coli*, using recombinant DNA technology, is identical to the 34 N-terminal amino acid sequence of endogenous human parathyroid hormone.

For the full list of excipients, see *section Excipients*.

DESCRIPTION:

Colourless, clear solution for injection.

KAULIV is a biosimilar medicinal product of Forteo.

INDICATION:

KAULIV is indicated in adults.

Treatment of osteoporosis in postmenopausal women and in men at increased risk of fracture (see *section Pharmacodynamic properties*). In postmenopausal women, a significant reduction in the incidence of vertebral and non- vertebral fractures but not hip fractures has been demonstrated.

Treatment of osteoporosis associated with sustained systemic glucocorticoid therapy in women and men at increased risk for fracture (see *section Pharmacodynamic properties*).

RECOMMENDED DOSAGE:

The recommended dose of KAULIV is 20 micrograms administered once daily.

The maximum total duration of treatment with KAULIV should be 24 months (see *section Warnings and precautions*). The 24-month course of KAULIV should not be repeated over a patient's lifetime.

Patients should receive supplemental calcium and vitamin D supplements if dietary intake is inadequate.

Following cessation of KAULIV therapy, patients may be continued on other osteoporosis therapies.

Special populations*Patients with renal impairment*

KAULIV must not be used in patients with severe renal impairment (see *section Contraindications*). In patients with moderate renal impairment, KAULIV should be used with caution. No special caution is required for patients with mild renal impairment.

Patients with hepatic impairment

No data are available in patients with impaired hepatic function (see *section Preclinical safety data*). Therefore, KAULIV should be used with caution.

Paediatric population and young adults with open epiphyses

The safety and efficacy of KAULIV in children and adolescents less than 18 years has not been established. KAULIV should not be used in paediatric patients (less than 18 years), or young adults with open epiphyses.

Elderly patients

Dosage adjustment based on age is not required (see *section Pharmacokinetics*).

Method of administration

KAULIV should be administered once daily by subcutaneous injection in the thigh or abdomen.

Patients must be trained to use the proper injection techniques (see *section Special precautions for disposal*). For instructions of the medicinal product before administration (see *section Special precautions for disposal*). Instructions for use which is included in the carton of the pen are also available to instruct patients on the correct use of the pen.

CONTRAINDICATIONS:

- Hypersensitivity to the active substance or to any of the excipients of this product.
- Pregnancy and breast-feeding (see *section Warnings and precautions* and *section Use in pregnancy and lactation*)
- Pre-existing hypercalcaemia
- Severe renal impairment
- Metabolic bone diseases (including hyperparathyroidism and Paget's disease of the bone) other than primary osteoporosis or glucocorticoid-induced osteoporosis.
- Unexplained elevations of alkaline phosphatase
- Prior external beam or implant radiation therapy to the skeleton
- Patients with skeletal malignancies or bone metastases should be excluded from treatment with teriparatide.

WARNINGS & PRECAUTIONS:**Traceability**

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Serum and urine calcium

In normocalcaemic patients, slight and transient elevations of serum calcium concentrations have been observed following teriparatide injection. Serum calcium concentrations reach a maximum between 4 and 6 hours and return to baseline by 16 to 24 hours after each dose of teriparatide. Therefore, if blood samples for serum calcium measurements are taken, this should be done at least 16 hours after the most recent KAULIV injection. Routine calcium monitoring during therapy is not required.

KAULIV may cause small increases in urinary calcium excretion, but the incidence of hypercalciuria did not differ from that in the placebo-treated patients in clinical trials.

Urolithiasis

KAULIV has not been studied in patients with active urolithiasis. KAULIV should be used with caution in patients with active or recent urolithiasis because of the potential to exacerbate this condition.

Orthostatic hypotension

In short-term clinical studies with teriparatide, isolated episodes of transient orthostatic hypotension were observed. Typically, an event began within 4 hours of dosing and spontaneously resolved within a few minutes to a few hours. When transient orthostatic hypotension occurred, it happened within the first several doses, was relieved by placing subjects in a reclining position, and did not preclude continued treatment.

Renal impairment

Caution should be exercised in patients with moderate renal impairment.

Younger adult population

Experience in the younger adult population, including premenopausal women, is limited (see section *Pharmacodynamics*). Treatment should only be initiated if the benefit clearly outweighs risks in this population.

Women of childbearing potential should use effective methods of contraception during use of KAULIV. If pregnancy occurs, KAULIV should be discontinued.

Duration of treatment

Studies in rats indicate an increased incidence of osteosarcoma with long-term administration of teriparatide (see section *Preclinical safety data*). Until further clinical data become available, the recommended treatment time of 24 months should not be exceeded.

Sodium content

This medicine contains less than 1 mmol sodium (23 mg) per dose, that is to say essentially 'sodium-free'.

INTERACTION WITH OTHER MEDICAMENTS:

In a study of 15 healthy subjects administered digoxin daily to steady state, a single teriparatide dose did not alter the cardiac effect of digoxin. However, sporadic case reports have suggested that hypercalcaemia may predispose patients to digitalis toxicity. Because teriparatide transiently increases serum calcium, teriparatide should be used with caution in patients taking digitalis.

Teriparatide has been evaluated in pharmacodynamic interaction studies with hydrochlorothiazide. No clinically significant interactions were noted.

Co-administration of raloxifene or hormone replacement therapy with teriparatide did not alter the effects of teriparatide on serum or urine calcium or on clinical adverse events.

USE IN PREGNANCY & LACTATION:**Women of childbearing potential / Contraception in females**

Women of childbearing potential should use effective methods of contraception during use of KAULIV. If pregnancy occurs, KAULIV should be discontinued.

Pregnancy

KAULIV is contraindicated for use during pregnancy (see section *Contraindications*).

Breast-feeding

KAULIV is contraindicated for use during breast-feeding. It is not known whether teriparatide is excreted in human milk.

Fertility

Studies in rabbits have shown reproductive toxicity (see section *Preclinical safety data*). The effect of teriparatide on human foetal development has not been studied. The potential risk for humans is unknown.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

KAULIV has no or negligible influence on the ability to drive and use machines. Transient, orthostatic hypotension or dizziness was observed in some patients. These patients should refrain from driving or the use of machines until symptoms have subsided.

SIDE EFFECTS:**Summary of the safety profile**

The most commonly reported adverse reactions in patients treated with teriparatide are nausea, pain in limb, headache and dizziness.

Tabulated list of adverse reactions

Of patients in the teriparatide trials, 82.8% of the teriparatide patients and 84.5% of the placebo patients reported at least 1 adverse event.

The adverse reactions associated with the use of teriparatide in osteoporosis clinical trials and post-marketing exposure are summarised in the Table 1 below. The following convention has been used for the classification of the adverse reactions: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$) very rare ($< 1/10,000$).

Table 1. Adverse reactions

MedDRA system organ class	Frequency	Adverse reactions
Blood and lymphatic system disorders	Common	Anaemia
Immune system disorders	Rare	Anaphylaxis
Metabolism and nutrition disorders	Common	Hypercholesterolaemia
	Uncommon	Hypercalcaemia greater than 2.76 mmol/L, hyperuricaemia
	Rare	Hypercalcaemia greater than 3.25 mmol/L
Psychiatric disorders	Common	Depression
Nervous system disorders	Common	Dizziness, headache, sciatica, syncope
Ear and labyrinth disorders	Common	Vertigo
Cardiac disorders	Common	Palpitations
	Uncommon	Tachycardia
Vascular disorders	Common	Hypotension
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea
	Uncommon	Emphysema
Gastrointestinal disorders	Common	Nausea, vomiting, hiatus hernia, gastroesophageal reflux disease
	Uncommon	Haemorrhoids
Skin and subcutaneous tissue disorders	Common	Sweating increased
Musculoskeletal and connective tissue disorders	Very common	Pain in limb
	Common	Muscle cramps
	Uncommon	Myalgia, arthralgia, back cramp/pain*
Renal and urinary disorders	Uncommon	Urinary incontinence, polyuria, micturition urgency, nephrolithiasis
	Rare	Renal failure/impairment
General disorders and administration site condition	Common	Fatigue, chest pain, asthenia, mild and transient injection site events, including pain, swelling, erythema, localised bruising, pruritus and minor bleeding at injection site
	Uncommon	Injection site erythema, injection site reaction
	Rare	Possible allergic events soon after injection: acute dyspnoea, oro/facial oedema, generalised urticaria, chest pain, oedema (mainly peripheral)
Investigations	Uncommon	Weight increased, cardiac murmur, alkaline phosphatase increased

*Serious cases of back cramp or pain have been reported within minutes of the injection.

Description of selected adverse reactions

In clinical trials the following reactions were reported at a $\geq 1\%$ difference in frequency from placebo: vertigo, nausea, pain in limb, dizziness, depression, dyspnoea.

Teriparatide increases serum uric acid concentrations. In clinical trials, 2.8% of teriparatide patients had serum uric acid concentrations above the upper limit of normal compared with 0.7% of placebo patients. However, the hyperuricemia did not result in an increase in gout, arthralgia, or urolithiasis.

In a large clinical trial, antibodies that cross-reacted with teriparatide were detected in 2.8% of women receiving teriparatide. Generally, antibodies were first detected following 12 months of treatment and diminished after withdrawal of therapy. There was no evidence of hypersensitivity reactions, allergic reactions, effects on serum calcium, or effects on Bone Mineral Density (BMD) response.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Signs and symptoms

Teriparatide has been administered in single doses of up to 100 micrograms and in repeated doses of up to 60 micrograms/day for 6 weeks.

The effects of overdose that might be expected include delayed hypercalcaemia and risk of orthostatic hypotension. Nausea, vomiting, dizziness, and headache can also occur.

Overdose experience based on post-marketing spontaneous reports

In post-marketing spontaneous reports, there have been cases of medication error where the entire contents (up to 800 micrograms) of the teriparatide pen have been administered as a single dose. Transient events reported have included nausea, weakness/lethargy and hypotension. In some cases, no adverse events occurred as a result of the overdose. No fatalities associated with overdose have been reported.

Overdose management

There is no specific antidote for teriparatide. Treatment of suspected overdose should include transitory discontinuation of teriparatide, monitoring of serum calcium, and implementation of appropriate supportive measures, such as hydration.

PHARMACODYNAMICS:

Pharmacotherapeutic group: Calcium homeostasis, parathyroid hormones and analogues, ATC code: H05AA02.

Mechanism of action

Endogenous 84-amino-acid parathyroid hormone (PTH) is the primary regulator of calcium and phosphate metabolism in bone and kidney. Teriparatide (rhPTH(1-34)) is the active fragment (1-34) of endogenous human parathyroid hormone. Physiological actions of PTH include stimulation of bone formation by direct effects on bone forming cells (osteoblasts) indirectly increasing the intestinal absorption of calcium and increasing the tubular re-absorption of calcium and excretion of phosphate by the kidney.

Pharmacodynamic effects

Teriparatide is a bone formation agent to treat osteoporosis. The skeletal effects of teriparatide depend upon the pattern of systemic exposure. Once-daily administration of teriparatide increases apposition of new bone on trabecular and cortical bone surfaces by preferential stimulation of osteoblastic activity over osteoclastic activity.

Clinical study of KAULIV: Comparative pharmacokinetic study

The pharmacokinetic and pharmacodynamic effects of KAULIV were compared with reference products Forteo (United States reference product) and Forsteo (European Union reference product) during a double-blind, randomized, three-treatment, three-period, single-dose, cross-over study in healthy men and non-pregnant women after subcutaneous administration. In this study, 78 subjects were randomized to receive single subcutaneous dose of 20 micrograms each of KAULIV, Forteo and Forsteo in three periods separated by a washout period of at least one week between injections in each treatment period. Primary objective was to evaluate the pharmacokinetic equivalence of KAULIV against Forteo and Forsteo after single subcutaneous injection of 20 micrograms. Maximum measured concentration (C_{max}), area under plasma concentration versus time curve till the last time point (AUC_{0-t}), and area under plasma concentration versus time curve extrapolated to the infinity (AUC_{0-inf}) were primary pharmacokinetic parameters. The secondary objectives were to evaluate safety as well as tolerability, and to evaluate the transient effects of KAULIV against Forteo and Forsteo on baseline-corrected ionized serum calcium concentration after single subcutaneous injection of 20 micrograms.

The summary of statistical comparisons for primary pharmacokinetic parameters of KAULIV against Forteo and Forsteo is provided in Table 2. The 90% confidence intervals of the geometric least square mean ratios of C_{max}, AUC_{0-t} and AUC_{0-inf} for the comparisons of KAULIV against Forteo and Forsteo were within 80.00% to 125.00%.

Table 2. Summary of Statistical Comparison of Primary Pharmacokinetic Parameters of KAULIV with Forteo and Forsteo

KAULIV versus Forteo				
Parameter	Geometric least square means			90% confidence interval
	Kauliv	Forteo	Test/Reference (%)	
C_{max}	110.1	104.3	105.55	99.83 – 111.60
AUC_{0-t}	117.9	108.3	108.80	101.73 – 116.36
AUC_{0-inf}	133.8	120.6	110.93	104.32 – 117.97
KAULIV versus Forsteo				
Parameter	Geometric least square means			90% confidence interval
	Kauliv	Forsteo	Test/Reference (%)	
C_{max}	110.6	110.1	100.40	95.20 – 105.87
AUC_{0-t}	117.3	106.4	110.26	103.77 – 117.16
AUC_{0-inf}	127.5	114.8	111.07	105.96 – 116.44

Clinical efficacy and safety for Forteo**Risk Factors**

Independent risk factors, for example, low BMD, age, the existence of previous fracture, family history of hip fractures, high bone turnover and low body mass index should be considered in order to identify women and men at increased risk of osteoporotic fractures who could benefit from treatment.

Premenopausal women with glucocorticoid-induced osteoporosis should be considered at high risk for fracture if they have a prevalent fracture or a combination of risk factors that place them at high risk for fracture (e.g., low bone density [e.g., T score ≤ -2], sustained high dose glucocorticoid therapy [e.g., ≥ 7.5 mg/day for at least 6 months], high underlying disease activity, low sex steroid levels).

Postmenopausal osteoporosis

The pivotal study included 1637 postmenopausal women (mean age 69.5 years). At baseline, ninety percent of the patients had one or more vertebral fractures, and on average, vertebral BMD was 0.82 g/cm² (equivalent to a T-score = -2.6). All patients were offered 1000 mg calcium per day and at least 400 IU vitamin D per day. Results from up to 24 months (median: 19 months) treatment with teriparatide demonstrate statistically significant fracture reduction (Table 3). To prevent one or more new vertebral fractures, 11 women had to be treated for a median of 19 months.

Table 3. Fracture incidence in postmenopausal women

	Placebo (N = 544) (%)	Teriparatide (N = 541) (%)	Relative risk (95 % CI) vs. placebo
New vertebral fracture (≥ 1) ^a	14.3	5.0 ^b	0.35 (0.22, 0.55)
Multiple vertebral fractures (≥ 2) ^a	4.9	1.1 ^b	0.23 (0.09, 0.60)
Non-vertebral fragility fractures ^c	5.5	2.6 ^d	0.47 (0.25, 0.87)

Major non-vertebral fragility fractures ^c (hip, radius, humerus, ribs and pelvis)	3.9	1.5 ^d	0.38 (0.17, 0.86)
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Abbreviations: N = number of patients randomly assigned to each treatment group; CI = Confidence Interval.

^a The incidence of vertebral fractures was assessed in 448 placebo and 444 teriparatide patients who had baseline and follow-up spine radiographs.

^b $p \leq 0.001$ compared with placebo

^c A significant reduction in the incidence of hip fractures has not been demonstrated

^d $p \leq 0.025$ compared with placebo.

After 19 months (median) treatment, bone mineral density (BMD) had increased in the lumbar spine and total hip, respectively, by 9% and 4% compared with placebo ($p < 0.001$).

Post-treatment management: Following treatment with teriparatide, 1262 postmenopausal women from the pivotal trial enrolled in a post-treatment follow-up study. The primary objective of the study was to collect safety data of teriparatide. During this observational period, other osteoporosis treatments were allowed and additional assessment of vertebral fractures was performed.

During a median of 18 months following discontinuation of teriparatide, there was a 41% reduction ($p = 0.004$) compared with placebo in the number of patients with a minimum of one new vertebral fracture. In an open-label study, 503 postmenopausal women with severe osteoporosis and a fragility fracture within the previous 3 years (83 % had received previous osteoporosis therapy) were treated with teriparatide for up to 24 months. At 24 months, the mean increase from baseline in lumbar spine, total hip and femoral neck BMD was 10.5 %, 2.6 % and 3.9 % respectively. The mean increase in BMD from 18 to 24 months was 1.4 %, 1.2 % and 1.6 % at the lumbar spine, total hip and femoral neck, respectively.

A 24-month, randomized, double-blind, comparator-controlled Phase 4 study included 1,360 postmenopausal women with established osteoporosis. 680 subjects were randomised to teriparatide and 680 subjects were randomised to oral risendronate 35 mg/week. At baseline, the women had a mean age of 72.1 years and a median of 2 prevalent vertebral fractures; 57.9% of patients had received previous bisphosphonate therapy and 18.8% took concomitant glucocorticoids during the study. 1,013 (74.5%) patients completed the 24-month follow-up. The mean (median) cumulative dose of glucocorticoid was 474.3 (66.2) mg in the teriparatide arm and 898.0 (100.0) mg in the risendronate arm. The mean (median) vitamin D intake for the teriparatide arm was 1433 IU/day (1400 IU/day) and for the risendronate arm was 1191 IU/day (900 IU/day). For those subjects who had baseline and follow-up spine radiographs, the incidence of new vertebral fractures was 28/516 (5.4%) in teriparatide- and 64/533 (12.0%) in risendronate-treated patients, relative risk (95% CI) = 0.44 (0.29-0.68), $P < 0.0001$. The cumulative incidence of pooled clinical fractures (clinical vertebral and non-vertebral fractures) was 4.8% in teriparatide and 9.8% in risendronate-treated patients, hazard ratio (95% CI) = 0.48 (0.32-0.74), $P = 0.0009$.

Male osteoporosis

437 patients (mean age 58.7 years) were enrolled in a clinical trial for men with hypogonadal (defined as low morning free testosterone or an elevated FSH or LH) or idiopathic osteoporosis. Baseline spinal and femoral neck bone mineral density mean T-scores were -2.2 and -2.1, respectively. At baseline, 35% of patients had a vertebral fracture and 59% had a non-vertebral fracture.

All patients were offered 1000 mg calcium per day and at least 400 IU vitamin D per day. Lumbar spine BMD significantly increased by 3 months. After 12 months, BMD had increased in the lumbar spine and total hip by 5% and 1%, respectively, compared with placebo. However, no significant effect on fracture rates was demonstrated.

Glucocorticoid-induced osteoporosis

The efficacy of teriparatide in men and women (N=428) receiving sustained systemic glucocorticoid therapy (equivalent to 5 mg or greater of prednisone for at least 3 months) was demonstrated in the 18- month primary phase of a 36-month, randomised, double-blind, comparator-controlled study (alendronate 10 mg/day). Twenty-eight percent of patients had one or more radiographic vertebral fractures at baseline. All patients were offered 1000 mg calcium per day and 800 IU vitamin D per day.

This study included postmenopausal women (N=277), premenopausal women (N=67), and men (N=83). At baseline, the postmenopausal women had a mean age of 61 years, mean lumbar spine BMD T score of -2.7, median prednisone equivalent dose of 7.5 mg/day, and 34% had one or more radiographic vertebral fractures; premenopausal women had a mean age of 37 years, mean lumbar spine BMD T score of -2.5, median prednisone equivalent dose of 10 mg/day, and 9% had one or more radiographic vertebral fractures; and men had a mean age of 57 years, mean lumbar spine BMD T score of -2.2, median prednisone equivalent dose of 10 mg/day, and 24% had one or more radiographic vertebral fractures.

Sixty-nine percent of patients completed the 18-month primary phase. At the 18-month endpoint, teriparatide significantly increased lumbar spine BMD (7.2%) compared with alendronate (3.4%) ($p < 0.001$). Teriparatide increased BMD at the total hip (3.6%) compared with alendronate (2.2%) ($p < 0.01$), as well as at the femoral neck (3.7%) compared with alendronate (2.1%) ($p < 0.05$). In patients treated with teriparatide, lumbar spine, total hip and femoral neck BMD increased between 18 and 24 months by an additional 1.7 %, 0.9 % and 0.4 %, respectively.

At 36 months, analysis of spinal X-rays from 169 alendronate patients and 173 teriparatide patients showed that 13 patients in the alendronate group (7.7 %) had experienced a new vertebral fracture compared with 3 patients in the teriparatide group (1.7 %) ($p = 0.01$). In addition, 15 of 214 patients in the alendronate group (7.0 %) had experienced a non-vertebral fracture compared with 16 of 214 patients in the Teriparatide group (7.5 %) ($p = 0.84$).

In premenopausal women, the increase in BMD from baseline to 18-month endpoint was significantly greater in the teriparatide group compared with the alendronate group at the lumbar spine (4.2% versus -1.9%; $p < 0.001$) and total hip (3.8% versus 0.9%; $p = 0.005$). However, no significant effect on fracture rates was demonstrated.

PHARMACOKINETICS:**Distribution**

The volume of distribution is approximately 1.7 L/kg. The half-life of teriparatide is approximately 1 hour when administered subcutaneously, which reflects the time required for absorption from the injection site.

Biotransformation

No metabolism or excretion studies have been performed with teriparatide, but the peripheral metabolism of parathyroid hormone is believed to occur predominantly in liver and kidney.

Elimination

Teriparatide is eliminated through hepatic and extra-hepatic clearance (approximately 62 L/hr in women and 94 L/hr in men).

Special populations**Elderly**

No differences in teriparatide pharmacokinetics were detected with regard to age (range 31 to 85 years). Dosage adjustment based on age is not required.

PRECLINICAL SAFETY DATA

Teriparatide was not genotoxic in a standard battery of tests. It produced no teratogenic effects in rats, mice or rabbits. There were no important effects observed in pregnant rats or mice administered teriparatide at daily doses of 30 to 1000 µg/kg. However, fetal resorption and reduced litter size occurred in pregnant rabbits administered daily doses of 3 to 100 µg/kg. The embryotoxicity observed in rabbits may be related to their much greater sensitivity to the effects of PTH on blood ionised calcium compared with rodents.

Rats treated with near-life time daily injections had dose-dependent exaggerated bone formation and increased incidence of osteosarcoma most probably due to an epigenetic mechanism. Teriparatide did not increase the incidence of any other type of neoplasia in rats. Due to the differences in bone physiology in rats and humans, the clinical relevance of these findings is probably minor. No bone tumours were observed in ovariectomised monkeys treated for 18 months or during a 3-year follow-up period after treatment cessation. In addition, no osteosarcomas have been observed in clinical trials or during the post treatment follow-up study.

Animal studies have shown that severely reduced hepatic blood flow decreases exposure of PTH to the principal cleavage system (Kupffer cells) and consequently clearance of PTH (1-84).

EXCIPIENTS

D-Mannitol

Metacresol

Acetic acid glacial

Sodium acetate anhydrous

Diluted hydrochloric acid (for pH adjustment)

Sodium hydroxide (for pH adjustment)

Water for injection

INCOMPATIBILITIES

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

SHELF LIFE:

24 months.

Refer to the expiry date on the product labels.

Chemical, physical, and microbiological in-use stability has been demonstrated for 28 days at 2 - 8°C. Once opened, the cartridge may be stored for a maximum of 28 days at 2 - 8°C. After insertion of the cartridge into the pen, the combined pen and cartridge should be returned to the refrigerator immediately after use. Discard the cartridge 28 days after the first use.

Do not store the pen with the needle attached. Do not remove the cartridge from the pen after first use. The cartridge within the pen can additionally be placed in the pouch supplied with the pen in order to protect from light

Other in-use storage times and conditions are the responsibility of the user.

STORAGE CONDITIONS:

Store in a refrigerator (2°C - 8°C) at all times.

Do not freeze. Keep the cartridge in the outer carton in order to protect from light.

For storage conditions after first opening of the medicinal product, see section *Shelf life*.

NATURE AND CONTENTS OF CONTAINER

3 mL cartridge (USP type-1 glass cartridge), with a plunger stopper (bromobutyl) and disc seal (aluminium and rubber liner seals), packed in a plastic tray sealed with lid foil and supplied in a carton.

Each cartridge contains 3 mL solution for injection corresponding to 28 doses of 20 micrograms (per 80 microliters).

Pack sizes

KAULIV 1 cartridge or 3 cartridges.

Not all pack sizes may be marketed.

SPECIAL PRECAUTIONS FOR DISPOSAL**Handling**

KAULIV cartridges should be used exclusively within KAULIV reusable, multidose pen. No needles are supplied with this medicinal product.

Each cartridge and pen should be used by only one patient. The pen can be used with 32 G 4 mm single-use pen needles.

A new, sterile needle must be used for every injection.

The expiry date on the cartridge label must always be checked before inserting the cartridge into KAULIV pen. To avoid medication errors, make sure that the date when starting to use a new cartridge is at least 28 days before its expiry date.

Before using the pen for the first time, the patient should read and understand the instructions on how to use the pen, which are provided with the pen.

After each injection, the pen should be returned to the refrigerator. After the first use, the cartridge should not be removed from the pen during the 28 days of usage. Do not use KAULIV if it is, or has been frozen.

KAULIV must not be transferred to a syringe. Empty cartridges must not be refilled.

KAULIV should not be used if the solution is cloudy, coloured or contains visible particles.

The date of first injection should be written on the outer carton of KAULIV cartridge (see the provided space on the box: “First use”).

The KAULIV reusable pen features a dose selector with audible clicks and visual indicators to ensure correct dose is set for priming (P) and for setting the dose (D).

Disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

MANUFACTURER AND RELEASED BY:

OneSource Specialty Pharma Limited
D1 Plot No. 2, A Taluk, Doddaballapur,
Karnataka – 561 203, India.

PRODUCT REGISTRATION HOLDER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, Malaysia.

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PLEASE FOLLOW THESE STEP-BY-STEP INSTRUCTIONS CAREFULLY BEFORE USING KAULIV PEN.

THE DEVICE SHOULD NOT BE USED FOR ANY OTHER PURPOSE THAN THAT ADVISED BY YOUR HEALTHCARE PROFESSIONAL.

FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN AN INACCURATE DOSE.

INDICATIONS FOR USE

Kauliv pen is a reusable injection pen for self-administering subcutaneous injections of teriparatide in the treatment of osteoporosis. For further information, please refer to "What Kauliv is and what it is used for" in the package leaflet for Kauliv.

The Kauliv pen must only be used with Strides 3 mL Kauliv cartridge and 4mm/32G single use pen needle.

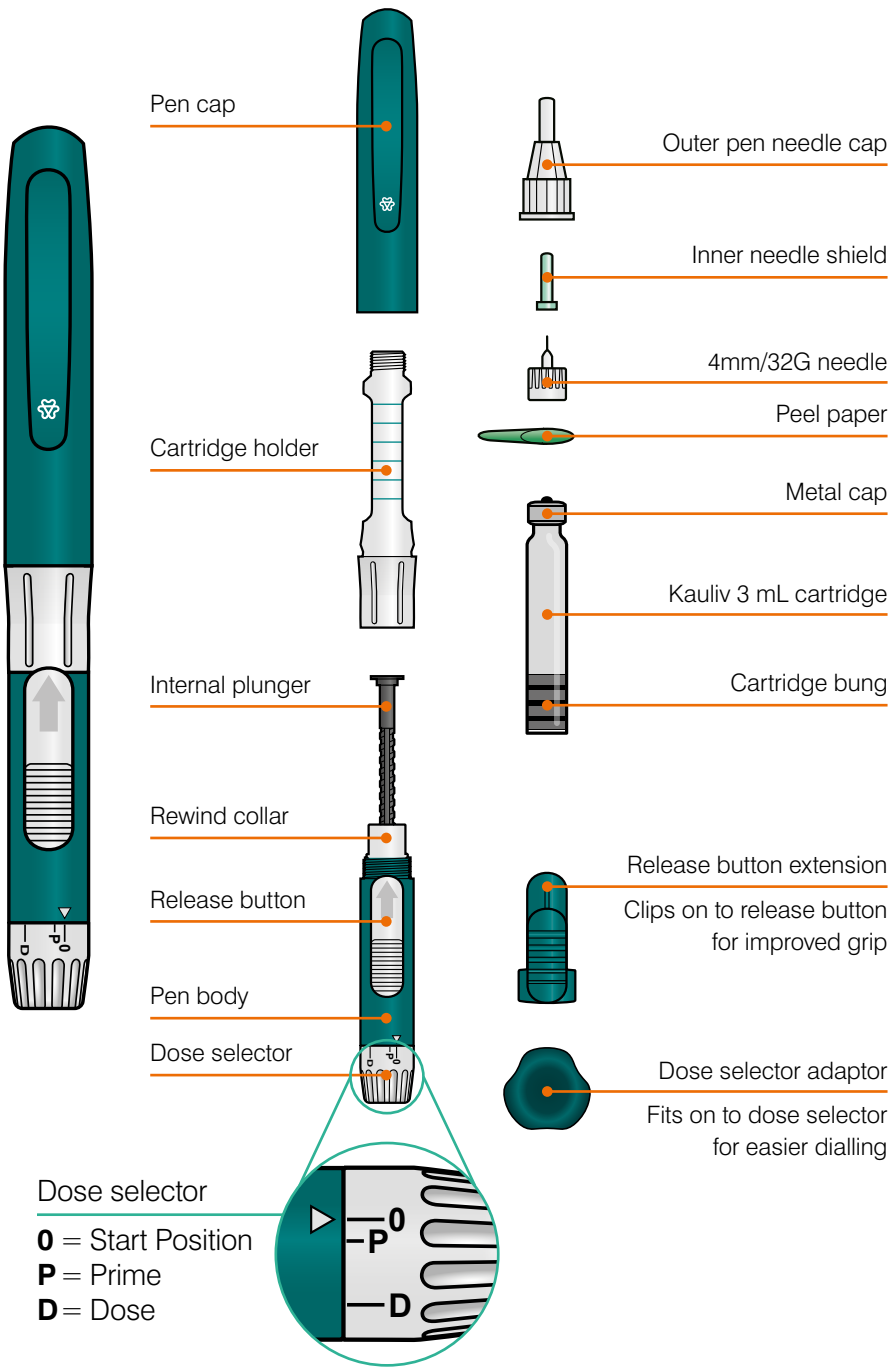
Each cartridge contains 28 doses of teriparatide. Inject a single dose (D) of 20 micrograms (corresponding to 80 microliters) every day.

You need to load a new cartridge at the start of each new 28 day period.

Patients and caregivers who administer Strides [teriparatide (rDNA origin) injection] should receive appropriate training and instruction on the proper use of the Kauliv pen from a qualified health professional. It is important to read, understand, and follow the instructions for using the pen as given in the Instructions for Use for Kauliv pen.

SAFETY PRECAUTIONS

- Wash hands thoroughly before use.
- When a Kauliv cartridge is loaded, the Kauliv pen should be stored in a refrigerator (2°C to 8°C) in between injections until 28 days of use from the date of first use.
- The Kauliv pen should be stored with the arrow ► on the pen body pointing to 0 to avoid damage to the device.
- Check the Kauliv cartridge. If the solution appears cloudy, then use a new cartridge.
- Always ensure the Kauliv pen has been primed before every use, in accordance with the instructions detailed in steps 6 and 7. Failure to follow the exact priming procedures could result in an inaccurate dose being delivered.
- Never attempt to dial back the dose selector as doing so may damage the Kauliv pen. Expel the incorrect dose into the air.
- Do not inject more than one dose (D) per day.
- If when removing the pen needle any leakage of teriparatide is found, it is possible that you may not have injected the full teriparatide dose. Do not try to make up for the shortfall in your teriparatide dose by giving a second injection.
- The needle should be removed and safely disposed of immediately after each injection. If the needle is left on the pen, it can get blocked and affect the next dose.
- Choking hazard – small parts. Keep away from children under 3.



STORAGE AND CLEANING

- To clean the Kauliv pen wipe with a damp cloth. Do not immerse in water. Do not use any other solutions to clean the device, e.g. chemicals, alcohol wipes.
- The Kauliv pen should always be stored or carried with the pen needle removed and the pen cap on.
- Discard cartridge 28 days after the date of first use.
- Before disposing of the pen injector, be sure to remove the pen needle.
- Dispose of used pen needles carefully into a sharps container or in accordance with the advice given by your healthcare professional/local regulations.
- Do not throw needles directly into your household waste.
- Do not recycle the filled sharps container.
- Dispose of the pen following your local medical waste disposal procedure after 2 years from the date of first use.

LOAD A CARTRIDGE

1

Pull off the pen cap and **unscrew** the pen body from the cartridge holder (fig. 1).

2

Check to ensure that the internal plunger is fully retracted (fig. 2). If the internal plunger is extended, please refer to step 12 (fig. 12) to see how to wind it back in.

3

Check the teriparatide cartridge. If the solution appears cloudy, then use a new cartridge. Load the teriparatide cartridge into the cartridge holder, metal cap in first (fig. 3).

4

Screw the cartridge holder and pen body firmly together (fig. 4).

ATTACH A NEW NEEDLE

Peel the protective paper from a new pen needle (Size: 4mm/32G).

5a

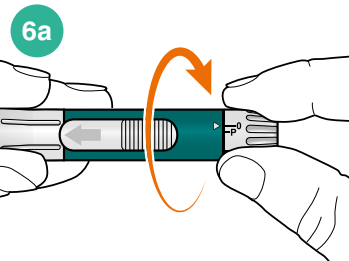
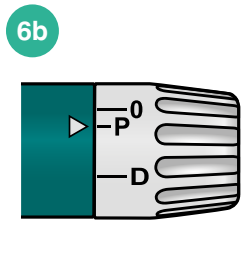
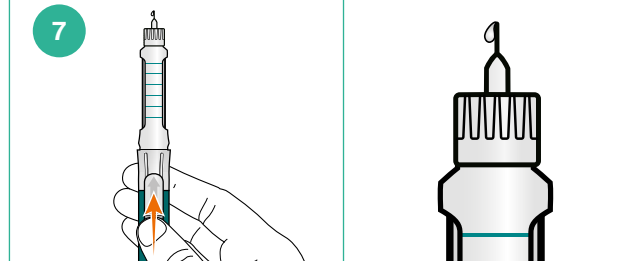
Screw the pen needle straight onto the cartridge holder (fig. 5a). The pen needle must be changed for every injection.

5b

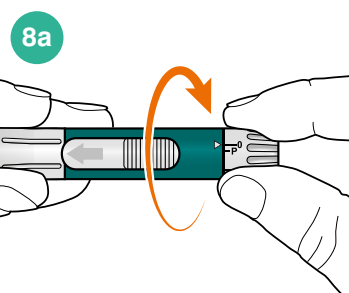
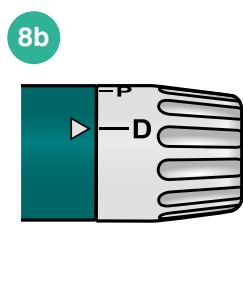
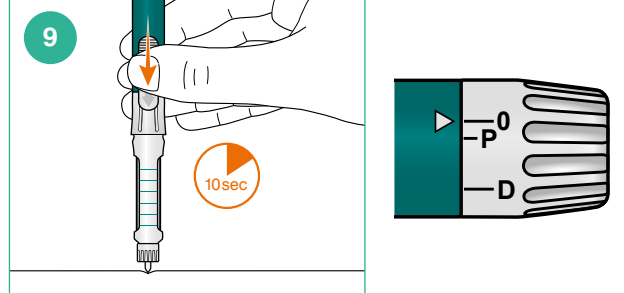
Pull off the outer pen needle cap and inner needle shield to expose the pen needle (fig. 5b).

Please turn over page

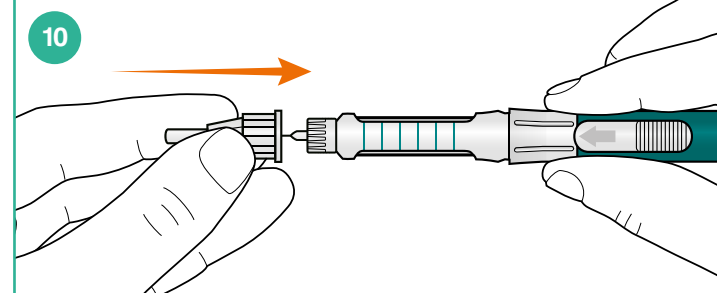
PRIME THE PEN

		
It is important to prime the Kauliv pen before every use to remove any air that may be inside the needle and cartridge. Dial the dose selector to P to prime the device (fig. 6b). If you over-dial, proceed to step 7 (fig. 7).		Hold the Kauliv pen with the needle pointing up. Push and hold the release button towards the needle until the arrow points to 0. Keep dialling to P and releasing until a drop of teriparatide drug appears at the needle tip (fig. 7). The Kauliv pen is now ready for use. (If the arrow ► does not return to 0 after pushing the release button, refer to the Priming advice given below.)

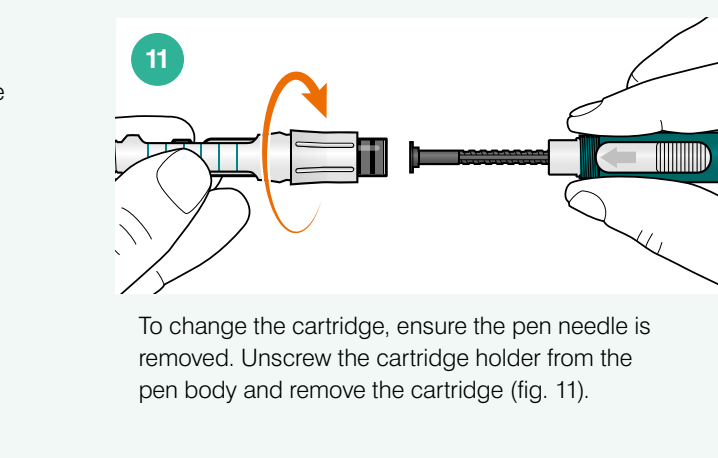
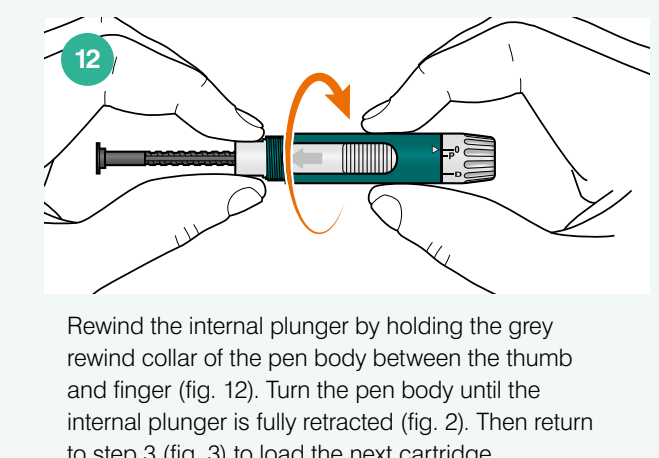
INJECT A DOSE

		
Make sure the arrow ► points to 0 on the dose selector (fig. 8a). Now dial up to D (fig. 8b).		Using the technique recommended by your healthcare professional gently insert the needle of the Kauliv pen into the injection site. Push the release button towards the pen needle and hold it in place until the arrow ► points to 0 and continue holding the release button down for 10 seconds, then remove the needle from the skin (fig. 9).

REMOVE THE NEEDLE


Reattach the outer pen needle cap and unscrew the pen needle (fig. 10). Always check your pen needle has been removed. Dispose of used pen needles carefully into an approved sharps container in accordance with the advice given by your healthcare professional or local regulations. Replace the pen cap on to the Kauliv pen and return it to the fridge for storage.

CHANGE THE CARTRIDGE (EVERY 28 DAYS)

Each cartridge contains enough teriparatide for 28 doses. After 28 days, dispose of the used cartridge and insert a new cartridge (steps 1 to 4). Dispose of the used cartridge in accordance with the advice given by your healthcare professional/local regulations.	 To change the cartridge, ensure the pen needle is removed. Unscrew the cartridge holder from the pen body and remove the cartridge (fig. 11).	 Rewind the internal plunger by holding the grey rewind collar of the pen body between the thumb and finger (fig. 12). Turn the pen body until the internal plunger is fully retracted (fig. 2). Then return to step 3 (fig. 3) to load the next cartridge.
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PRIMING ADVICE

If during step 7 (fig. 7) the arrow ► does not point to **0** (after pushing the release button) and no teriparatide appears, this may indicate one of two possibilities:

- A.** The pen needle that you are using could be blocked. If so, remove the pen needle from the pen and replace with a new one. Then return to steps and figures 5, 6 and 7.
- B.** The cartridge bung may require releasing. If so, follow these steps:
 - a.** Replace the outer pen needle cap onto the needle.
 - b.** Unscrew the cartridge holder.

- c.** Push and hold down the release button towards the internal plunger until the arrow ► points to **0** on the dose selector.
- d.** Dial up to **D** without screwing the device back together. Push and hold down the release button towards the internal plunger until the arrow ► points to **0**.
- e.** Without rewinding the internal plunger, screw the cartridge holder and body firmly together. This will release the cartridge bung and expel some teriparatide, however, the device will **not** be primed.
- f.** Repeat steps 6 and 7 to prime the device fully (fig. 6 and 7).

DEVICE WARRANTY

The Kauliv pen is under warranty for 2 years from date of purchase against any defective parts or workmanship providing there is no evidence of misuse. In the unlikely event of a failure, the product should be returned to your supplier.

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16 39